YEAR 10 PHYSICS, AQA GCSE, LESSON 3 OF 8

PARTICLE MODEL OF MATTER

LESSON 03

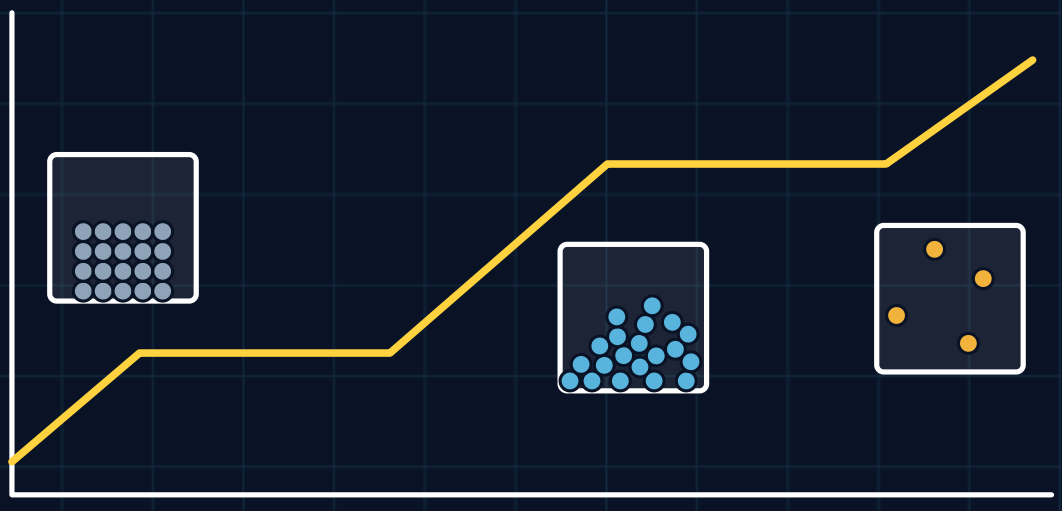
DENSITY

A gold bar the size of a carton of milk is too heavy for most people to lift with one hand. Why?

IN THIS LESSON YOU WILL

- calculate density with $\rho = \frac{m}{V}$
- use the units kg/m³ and g/cm³
- explain differences in density using the particle model
- rearrange the equation to find a mass or a volume

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME CLASS

START HERE

HOW THIS BOOKLET WORKS

This is the booklet for **Lesson 3** of the Particle Model of Matter unit. It builds on the lessons before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep the booklet: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
The explanation, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own, then a challenge.

PRACTICAL
Record your measurements and work out results from them.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an “I can” check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: The particles in a solid are close together **(close together / far apart)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of mass is **the newton** | **the kilogram** | **the joule** .

THE ROUTE THROUGH THE UNIT (THIS LESSON IS HIGHLIGHTED)

01 Solids, liquids and gases

02 Gas particles and pressure

03 Density
$\rho = \frac{m}{V}$, units, density and the particle model, finding a mass or a volume (slides 13–17)

04 Measuring density (required practical)

05 Internal energy and heating curves

06 Specific latent heat

07 Specific heat capacity

08 Measuring specific heat capacity (required practical)

The **equation toolkit** near the back has every equation you have met so far, with formula triangles to help you rearrange. Teaching deck: **panphy.app/gcsephy/decks/particle-model**

Lesson

03 DENSITY

AQA Physics 4.3.1.1; Combined Science 6.3.1.1

BIG QUESTION **A gold bar the size of a carton of milk is too heavy for most people to lift with one hand. Why?**

BY THE END YOU CAN

- calculate density with $\rho = \frac{m}{V}$
- use the units kg/m³ and g/cm³
- explain differences in density using the particle model
- rearrange the equation to find a mass or a volume

density **mass** **volume** **kilogram per cubic metre (kg/m³)** **rho (ρ)**

YOU WILL NEED

Per pair: calculator. Demonstration: blocks of the same size made of different materials (wood, aluminium, steel) to pass round.

Teaching deck, slides 13–17
panphy.app/gcsephy/decks/particle-model/#13

REMEMBER

DO NOW

From earlier lessons. No notes!

1
Describe the arrangement of the particles in a solid.

Very close together, in a regular pattern.

2
A gas in a sealed can is heated. What happens to its pressure?

It increases.

3
Convert 250 g to kilograms.

0.250 kg

EXPLAIN

HOW MUCH MASS IS PACKED IN?

Pick up a block of steel and a block of wood of the same size. The steel has much more mass, because more mass is packed into the same volume. We say that steel has a greater **density**.

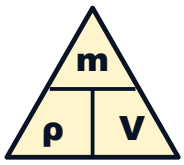
! **Density** is the mass of each unit of volume: the mass of 1 m³ (or of 1 cm³) of the material.

$$\rho = \frac{m}{V}$$

density = mass ÷ volume

ρ in kg/m³, m in kg, V in m³

Rearranged: $m = \rho V$ and $V = \frac{m}{\rho}$. The symbol ρ is the Greek letter rho.



WORKED EXAMPLE

A block of aluminium has a mass of 5400 kg and a volume of 2.0 m³. Calculate its density.

KNOW	m = 5400 kg, V = 2.0 m³
EQUATION	$\rho = \frac{m}{V}$
SUBSTITUTE	$\rho = \frac{5400}{2.0}$
ANSWER	$\rho = 2700$ kg/m³

YOUR TURN

A tank holds 3.0 m³ of water. The mass of the water is 3000 kg. Calculate the density of water.

KNOW	**m = 3000 kg, V = 3.0 m³**
EQUATION	$\rho = \frac{m}{V}$
SUBSTITUTE	$\rho = \frac{3000}{3.0}$
ANSWER	$\rho = 1000$ kg/m³

TWO SETS OF UNITS

In the lab it is easier to measure mass in grams and volume in cubic centimetres. Then the density is in **g/cm³**. Use one set of units or the other; never mix them.

MATERIAL	DENSITY IN G/CM³	DENSITY IN KG/M³
air	0.0012	1.2
ice	0.92	920
water	1.0	1000
aluminium	2.7	2700
iron	7.9	7900
gold	19.3	19 300

Get the marks: To change g/cm³ into kg/m³, **multiply by 1000**. To change kg/m³ into g/cm³, divide by 1000. Water is the one to remember: 1.0 g/cm³ = 1000 kg/m³.

DENSITY AND THE PARTICLE MODEL

In a solid and in a liquid the particles are close together, so a lot of mass fits into each cubic centimetre: solids and liquids have high densities. In a gas the particles are far apart, with empty space between them, so a gas has a very low density. Water is about 800 times denser than air.

Most substances are a little denser as a solid than as a liquid, because the particles are packed slightly closer. Water is unusual: ice is *less* dense than liquid water, which is why ice floats.

FINDING THE MASS OR THE VOLUME

WORKED EXAMPLE

The density of iron is 7900 kg/m³.
Calculate the mass of 0.020 m³ of iron.

KNOW	$\rho = 7900$ kg/m³, V = 0.020 m³
REARRANGE	$m = \rho V$
SUBSTITUTE	$m = 7900 \times 0.020$
ANSWER	**m = 158 kg**

YOUR TURN

The density of gold is 19.3 g/cm³.
Calculate the volume of a 965 g gold bar.

KNOW	$\rho = 19.3$ g/cm³, m = 965 g
REARRANGE	$V = \frac{m}{\rho}$
SUBSTITUTE	$V = \frac{965}{19.3}$
ANSWER	V = 50 cm³

PRACTISE

TRY IT

1 Draw a ring around the correct answer in each box.

The unit of density is | kg | (kg/m³) | m³/kg |

A gas has a low density because its particles are

very light	(far apart)	moving quickly

2 SUPPORTED A stone has a mass of 150 g and a volume of 60 cm³. Calculate its density in g/cm³.

EQUATION	$\rho = \frac{m}{V}$	
SUBSTITUTE	$\rho = \frac{150}{60}$	
ANSWER	**2.5 g/cm³**	unit:

3 SUPPORTED A concrete beam has a mass of 4800 kg and a volume of 2.0 m³. Calculate the density of the concrete.

EQUATION	$\rho = \frac{m}{V}$	
SUBSTITUTE	$\rho = \frac{4800}{2.0}$	
ANSWER	**2400 kg/m³**	unit:

4 Complete the table. Show your working underneath.

Object	Mass	Volume	Density
Wooden block	360 g	600 cm³	0.60 g/cm³
Copper pipe	445 g	50 cm³	8.9 g/cm³
Air in a room	60 kg	50 m³	1.2 kg/m³
Petrol in a tank	36 kg	0.050 m³	720 kg/m³

5 The density of sea water is 1.03 g/cm³. Write this density in kg/m³.

1.03 × 1000 = 1030 kg/m³

6 Explain, using the particle model, why 1 cm³ of liquid water has a much greater mass than 1 cm³ of steam.

In liquid water the particles are close together, but in steam they are far apart. So 1 cm³ of water contains many more particles, and more mass, than 1 cm³ of steam.

7 **CHALLENGE** A crown has a mass of 1.9 kg and a volume of 120 cm³. The density of gold is 19.3 g/cm³. Is the crown pure gold? Show your working. *Convert the mass to grams first.*

m = 1900 g. $\rho = \frac{m}{V} = \frac{1900}{120} = 15.8$ **g/cm³. This is less than 19.3 g/cm³, so the crown is not pure gold.**

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AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

> **Get the marks:** Write the equation in symbols first, then the numbers. A correct equation often earns a mark even if the arithmetic goes wrong.

8 A block of ice has a mass of 2760 kg and a volume of 3.0 m³. Calculate the density of ice. **[2 marks]**

EQUATION	$\rho = \frac{m}{V}$
SUBSTITUTE	$\rho = \frac{2760}{3.0}$ (1)
ANSWER	**920 kg/m³ (1)** unit:

9 A fuel tank holds 0.060 m³ of diesel. The density of diesel is 850 kg/m³. Calculate the mass of the diesel. **[3 marks]**

EQUATION	$\rho = \frac{m}{V}$
REARRANGE	$m = \rho V$ (1)
SUBSTITUTE	$m = 850 \times 0.060$ (1)
ANSWER	**51 kg (1)** unit:

10 Explain why a gas has a much lower density than a solid. **[2 marks]**

The particles in a gas are much further apart than in a solid (1), so there is less mass (fewer particles) in the same volume (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Density is the mass per unit volume: $\rho = \frac{m}{V}$.
- Units: kg/m³, or g/cm³ in the lab. 1 g/cm³ = 1000 kg/m³.
- Rearranged: $m = \rho V$ and $V = \frac{m}{\rho}$.
- Solids and liquids are dense because their particles are close together. Gases have low densities because their particles are far apart.

COVER THE PAGE AND ANSWER

1 Write the equation for density.

2 State the unit of density.

3 What is the density of water in kg/m³?

4 A block has a mass of 80 g and a volume of 20 cm³. What is its density?

I CAN...

- calculate density with $\rho = \frac{m}{V}$
- use kg/m³ and g/cm³ and convert between them
- rearrange the equation to find a mass or a volume
- explain differences in density using the particle model

REFERENCE

EQUATION TOOLKIT

Every equation from Lessons 1–3. AQA prints all of them on the equation sheet in the exam, but you still need to know what each symbol means, its unit, and how to use it.

1. Cover the quantity you want to find.
2. What is left side by side? **Multiply**.
3. One above the other? **Divide** the top by the bottom.

Example: to find V from $\rho = \frac{m}{V}$, cover V. m is above ρ, so $V = \frac{m}{\rho}$.

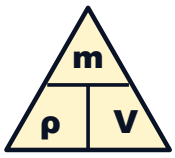
$$\rho = \frac{m}{V}$$

density = mass ÷ volume

kg/m³, kg, m³ (or g/cm³, g, cm³)

$m = \rho V$ and $V = \frac{m}{\rho}$

Lesson 3



1
Write what you know. Convert first: g → kg, cm³ → m³, kJ → J.

2
Write the equation in symbols.

3
Rearrange if you need to, then substitute the numbers.

4
Give the answer with a unit. Ask: is it sensible?

REFERENCE

KEY WORDS FROM THIS LESSON

Learn the meaning and the spelling. In exams, precise words earn marks: say “the particles vibrate about fixed positions”, not “the particles shake”.

Density (ρ): The mass of each unit of volume, in kg/m³ or g/cm³

Volume: The amount of space something takes up, in m³ or cm³

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. Very close together, in a regular pattern.
2. It increases.
3. 0.250 kg

COVER THE PAGE

1. $\rho = \frac{m}{V}$
2. kg/m³ (or g/cm³)
3. 1000 kg/m³
4. 4.0 g/cm³